

IMO 2026 Problem 1 — GCD/LCM Blackboard

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Abstract

There are 2026 integers greater than 1 on a blackboard. A move picks two entries $m, n > 1$ and replaces them with $\gcd(m, n)$ and $\text{lcm}(m, n) / \gcd(m, n)$. We prove that every play terminates after finitely many moves, that the terminal board contains exactly one entry $M > 1$, and that M is independent of the choices made during the play: explicitly, $M = \prod_p p^{G_p}$, where G_p is the gcd of the p -adic valuations of the *initial* entries. This “exponent-gcd number” is not, in general, the numeric gcd of the initial entries; we give counterexamples.

1 Introduction

This paper gives a complete solution of IMO 2026 Problem 1, the GCD/LCM blackboard problem. There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with

$$\gcd(m, n) \quad \text{and} \quad \frac{\text{lcm}(m, n)}{\gcd(m, n)}.$$

He continues to make moves while it is possible to do so.

- (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1.
- (b) Prove that the value of M does not depend on the choices of Confucius.

Two observations drive the solution. First, the identity $\text{lcm}(m, n) \cdot \gcd(m, n) = mn$ shows that the product of the two chosen entries drops by the factor $g = \gcd(m, n)$. Hence the product P of all board entries never increases, and it strictly decreases whenever $g > 1$. When $g = 1$ the pair becomes $(1, mn)$, so the number C of entries exceeding 1 drops by exactly one. The pair (P, C) therefore descends strictly in lexicographic order, and a move is possible exactly when $C \geq 2$; this proves termination and part (a).

Second, fix a prime p and read a move on the p -adic exponents of the two chosen entries: if the exponents are x, y , then the new entries have exponents $\min(x, y)$ and $|x - y|$, one step of the subtractive Euclidean algorithm. Since $\gcd(x, y) = \gcd(\min(x, y), |x - y|)$, the gcd G_p of the p -adic valuations of *all* entries is invariant under every move. At the terminal board every entry except M equals 1, so $v_p(M) = G_p$ for every prime p , and hence

$$M = \prod_p p^{G_p},$$

the *exponent-gcd number* of the initial multiset. This identifies M and proves part (b).

A word of caution: the formula $M = \prod_p p^{G_p}$ is *not* the numeric gcd of the initial entries in general; the invariant G_p is the gcd of the valuations, not their minimum. Counterexamples are collected in Section ??.

2 Statement of the Result

Theorem 1 (IMO 2026 Problem 1). *For the initial board of 2026 integers > 1 and the legal moves as defined:*

- (a) Every play terminates after finitely many moves, and the terminal board contains exactly one integer $M > 1$.
- (b) The value of M is independent of the choices made during the play; explicitly, $v_p(M) = G_p$ for every prime p , where $G_p = \gcd_i v_p(a_i)$ is the gcd of the p -adic valuations of the initial entries, so

$$M = \prod_p p^{G_p}$$

(the “exponent-gcd number” of the initial multiset), a finite product determined solely by the initial board.

The proof is given in Section ??; it rests on the lemmas of Section ?. The notation used in the statement is formalized in Section ?.

3 Notation and Preliminaries

For a positive integer n and a prime p , let $v_p(n)$ denote the p -adic valuation: the unique nonnegative integer such that $p^{v_p(n)} \mid n$ and $p^{v_p(n)+1} \nmid n$; set $v_p(1) = 0$. By the fundamental theorem of arithmetic, every positive integer n is uniquely $n = \prod_p p^{v_p(n)}$ (finitely many p with $v_p(n) > 0$); consequently two positive integers are equal iff their p -adic valuations agree for every prime p .

Every board arising in a play consists of positive integers: the initial entries exceed 1, and by Lemma ?? below each move replaces two positive integers by two positive integers. Hence valuations of entries are always defined.

For a finite multiset S of integers, define $\gcd(S)$ as the iterated gcd of its elements, with the conventions $\gcd(\emptyset) = 0$, $\gcd(0,0) = 0$, and $\gcd(x,0) = x$ for $x \geq 0$; the value is independent of the iteration order by associativity and commutativity of gcd.

4 Main Argument

4.1 Arithmetic of a Single Move

We record the elementary facts about one move.

Lemma 2 (Exponent formulas for gcd and lcm). *For a prime p and positive integers a, b ,*

$$v_p(\gcd(a, b)) = \min(v_p(a), v_p(b)), \quad v_p(\text{lcm}(a, b)) = \max(v_p(a), v_p(b)).$$

Proof. Write $a = \prod_q q^{\alpha_q}$, $b = \prod_q q^{\beta_q}$ with $\alpha_q = v_q(a)$, $\beta_q = v_q(b) \geq 0$. By unique factorization, $d \mid a$ and $d \mid b$ iff $v_q(d) \leq \min(\alpha_q, \beta_q)$ for every prime q ; hence the greatest common divisor of a, b has exponents $\delta_q = \min(\alpha_q, \beta_q)$, and, similarly, the least common multiple has exponents $\max(\alpha_q, \beta_q)$. Thus $v_p(\gcd(a, b)) = \min(v_p(a), v_p(b))$ and $v_p(\text{lcm}(a, b)) = \max(v_p(a), v_p(b))$ for each prime p . $\square$

Lemma 3 (Identity $\text{lcm} \cdot \gcd = mn$). *For positive integers m, n with $g = \gcd(m, n)$,*

$$\text{lcm}(m, n) \cdot \gcd(m, n) = mn, \quad \text{so} \quad \frac{\text{lcm}(m, n)}{\gcd(m, n)} = \frac{m}{g} \cdot \frac{n}{g} \in \mathbb{Z}_{>0}.$$

Under a legal move, the product P of all board entries changes from P to P/g , and the new pair $(g, \text{lcm}(m, n)/g)$ always contains at least one entry > 1 .

Proof. For each prime p , Lemma ?? gives $v_p(\text{lcm}(m, n) \gcd(m, n)) = \max(v_p(m), v_p(n)) + \min(v_p(m), v_p(n)) = v_p(mn)$, so the two positive integers are equal by unique factorization. Hence $\text{lcm}(m, n) / \gcd(m, n) = mn/g^2 = (m/g)(n/g)$, a positive integer because $g \mid m$ and $g \mid n$. The pair product is $g \cdot (mn/g^2) = mn/g$, so the total product P is divided by g . If $g > 1$, the first new entry $g > 1$; if $g = 1$, the new pair is $(1, mn)$ with $mn \geq 4$ (since $m, n \geq 2$). In both cases the new pair contains an entry > 1 . $\square$

Lemma 4 (Per-prime exponent dynamics of a move). *Fix a prime p . A legal move on entries m, n with $(x, y) = (v_p(m), v_p(n))$ replaces the exponent pair by $(\min(x, y), |x - y|)$. The p -valuations of all other entries are unchanged, and the same rule holds uniformly for every prime.*

Proof. By Lemma ??, $h = \text{lcm}(m, n) / \text{gcd}(m, n)$ is a positive integer, so $v_p(h)$ is well defined; writing $g = \text{gcd}(m, n)$, Lemma ?? gives

$$v_p(h) = v_p(\text{lcm}(m, n)) - v_p(g) = \max(x, y) - \min(x, y) = |x - y|.$$

The first new entry is g , with $v_p(g) = \min(x, y)$, and entries not moved keep their valuations. The argument uses only that p is prime, so it applies uniformly to every prime (for a prime dividing neither m nor n , the pair $(0, 0)$ is fixed). $\square$

Thus, for each prime p individually, a move performs one subtractive Euclidean step on the two chosen exponents and leaves all other exponents untouched.

4.2 Invariants of the Play

The following two lemmas isolate the preservation principle behind part (b): a move replaces a pair of exponents by a pair with the same gcd.

Lemma 5 (Euclidean step identity for gcd). *For all nonnegative integers x, y , with $\text{gcd}(0, 0) = 0$ and $\text{gcd}(x, 0) = x$,*

$$\text{gcd}(x, y) = \text{gcd}(\min(x, y), |x - y|).$$

Proof. Assume $x \leq y$; the other case is symmetric. Then $\min(x, y) = x$ and $|x - y| = y - x$. The ideals (x, y) and $(x, y - x)$ are equal: $y = (y - x) + x \in (x, y - x)$, and $y - x \in (x, y)$. In $\mathbb{Z}$ every ideal is principal, and the unique nonnegative generator of (a, b) is $\text{gcd}(a, b)$, so $\text{gcd}(x, y) = \text{gcd}(x, y - x)$. The conventions $\text{gcd}(0, 0) = 0$ and $\text{gcd}(x, 0) = x$ follow from $(0, 0) = 0\mathbb{Z}$ and $(x, 0) = x\mathbb{Z}$. $\square$

Lemma 6 (Multiset gcd preservation). *Let S be a finite multiset of integers with $|S| \geq 1$ and let S' be obtained from S by replacing two elements a, b at distinct positions by c, d such that $\text{gcd}(c, d) = \text{gcd}(a, b)$. Then $\text{gcd}(S') = \text{gcd}(S)$.*

Proof. Let R be the multiset obtained from S by deleting the two positions holding a and b . By the grouping identity for the iterated gcd, $\text{gcd}(S) = \text{gcd}(\text{gcd}(a, b), \text{gcd}(R))$ and $\text{gcd}(S') = \text{gcd}(\text{gcd}(c, d), \text{gcd}(R))$, which are equal by hypothesis. If $|S| = 2$ then $R = \emptyset$, and the convention $\text{gcd}(\emptyset) = 0$ together with $\text{gcd}(x, 0) = x$ covers this case. $\square$

Lemma 7 (Exponent-gcd invariant — part (b)). *For every prime p , the quantity*

$$G_p := \text{gcd}\{v_p(a) \mid a \text{ an entry of the board}\}$$

is invariant under every legal move, with the convention $\text{gcd}(0, 0) = 0$ on exponents.

Proof. At every board state all entries are positive integers, so $M_p(B) = \{v_p(a) : a \text{ on board}\}$ is a finite multiset of nonnegative integers and its iterated gcd $G_p(B)$ is well defined with the stated conventions.

Consider a legal move on entries m, n with $(x, y) = (v_p(m), v_p(n))$. By Lemma ??, the pair (x, y) is replaced by $(\min(x, y), |x - y|)$ and all other p -valuations are unchanged, so

$$M_p(B') = (M_p(B) \setminus \{x, y\}) \cup \{\min(x, y), |x - y|\}.$$

By Lemma ??, $\text{gcd}(\min(x, y), |x - y|) = \text{gcd}(x, y)$, so the replacement pair has the same gcd as the pair it replaces; applying Lemma ?? to $M_p(B)$ with $a = x, b = y, c = \min(x, y), d = |x - y|$ gives $G_p(B') = G_p(B)$.

Thus G_p is unchanged by every move; inductively over the move sequence, G_p is constant throughout any play and equals its initial value. $\square$

4.3 Termination

Part (a) rests on a lexicographic monovariant.

Lemma 8 (Termination — part (a)). *For any board of $k \geq 1$ integers > 1 (with $k = 2026$ in the problem) and any legal play:*

- (i) *The number C of entries > 1 satisfies $C \geq 1$ throughout and never increases.*

- (ii) The pair (P, C) , where P is the product of all entries, strictly decreases in lexicographic order after every move.
- (iii) Every play has at most $(k - 1) + \lfloor \log_2 P_0 \rfloor$ moves and is therefore finite.
- (iv) The terminal board has exactly one entry > 1 .

Proof. For a move on entries m, n with $g = \gcd(m, n)$, Lemma ?? gives that the new pair $(g, \text{lcm}(m, n)/g)$ has product mn/g and contains an entry > 1 .

Case $g > 1$: the total product is divided by $g \geq 2$, so P strictly decreases, and C does not increase (the new pair contributes at most two entries > 1).

Case $g = 1$: P is unchanged, and the new pair is $(1, mn)$ with $mn \geq 4 > 1$, so C decreases by exactly 1.

Hence C never increases; and $C \geq 1$ is preserved, because every new pair contains an entry > 1 (initially $C = k \geq 1$). Therefore (P, C) strictly decreases in lexicographic order after every move, proving (i) and (ii).

Moves with $g > 1$ divide P by at least 2 each, and $P \geq 1$ throughout, so there are at most $\lfloor \log_2 P_0 \rfloor$ of them. Moves with $g = 1$ decrease C by exactly 1, and $C \geq 1$, so there are at most $k - 1$ of them. Every play thus has at most $(k - 1) + \lfloor \log_2 P_0 \rfloor$ moves and is finite, proving (iii).

A move is possible exactly when $C \geq 2$ (two entries > 1 at distinct places); hence a maximal play stops only at $C = 1$, i.e. the terminal board has exactly one entry > 1 , proving (iv). $\square$

4.4 The Terminal Valuation

It remains to read the invariant G_p off the terminal board.

Lemma 9 (Terminal valuation equality). *At the terminal board of any legal play, with M the unique entry > 1 , for every prime p :*

$$v_p(M) = G_p,$$

where G_p is the invariant from Lemma ??, equal to the gcd of the p -adic valuations of the initial entries.

Proof. By Lemma ??(iv), the terminal board consists of M together with 2025 entries equal to 1. The exponent multiset at p is therefore $\{v_p(M), 0, \dots, 0\}$ (2025 zeros), since $v_p(1) = 0$, and its gcd is $v_p(M)$ by repeated use of $\gcd(x, 0) = x$ (the convention $\gcd(0, 0) = 0$ covers the case $v_p(M) = 0$). By Lemma ??, this terminal value equals the initial value of G_p , i.e. the gcd of the p -adic valuations of the initial 2026 entries. $\square$

5 Proof of the Main Theorem

The lemmas of Section ?? assemble into the proof of Theorem ??.

Proof of Theorem ??. Part (a). Instantiate Lemma ?? with $k = 2026$. All preconditions hold: there are 2026 entries all > 1 , a legal move selects two entries > 1 at distinct places, and play continues while possible. Hence every play is finite and its terminal board has exactly one entry $M > 1$.

Part (b). At the terminal board, the 2025 entries other than M are positive integers not exceeding 1, hence equal to 1. By Lemma ??, $v_p(M) = G_p$ for every prime p , where G_p is the gcd of the p -adic valuations of the initial entries. Unique prime factorization gives $M = \prod_p p^{v_p(M)} = \prod_p p^{G_p}$, a finite product: $G_p > 0$ only for primes dividing some initial entry (all other p -valuations are 0, so $G_p = \gcd(0, \dots, 0) = 0$ and $p^{G_p} = 1$). If two plays terminate at M_1 and M_2 , then $v_p(M_1) = G_p = v_p(M_2)$ for all p , so $M_1 = M_2$ by unique factorization; hence M is determined solely by the initial multiset and does not depend on the choices of Confucius.

Finally, $M > 1$: some initial entry is ≥ 2 and has a prime divisor p with $v_p(\text{that entry}) \geq 1$, so $G_p \geq 1$ and $p^{G_p} \mid M$, consistent with part (a). $\square$

6 Remarks

Remark 10 (M is not the numeric gcd of the initial entries). *The value $M = \prod_p p^{G_p}$ is in general **not** equal to the numeric gcd of the initial entries. Three examples:*

- *On the initial board $(4, 8)$, one possible play is $(4, 8) \rightarrow (4, 2) \rightarrow (2, 2) \rightarrow (2, 1)$, terminating with $M = 2$, while $\gcd(4, 8) = 4$. Indeed $G_2 = \gcd(v_2(4), v_2(8)) = \gcd(2, 3) = 1$, so $M = 2^1 = 2$.*
- *On the initial board $(2, 3)$, the only move gives $(2, 3) \rightarrow (1, 6)$, so $M = 6$, while $\gcd(2, 3) = 1$. Here $G_2 = \gcd(1, 0) = 1$ and $G_3 = \gcd(0, 1) = 1$, so $M = 2 \cdot 3 = 6$.*
- *On the initial board $(4, 8, 9)$, one has $G_2 = \gcd(2, 3, 0) = 1$ and $G_3 = \gcd(0, 0, 2) = 2$, hence $M = 2^1 \cdot 3^2 = 18$, while $\gcd(4, 8, 9) = 1$. The formula can produce an M much larger than the numeric gcd.*

The quantities differ because the p -adic valuation of the numeric gcd is $\min_i v_p(a_i)$, whereas $G_p = \gcd_i v_p(a_i)$ is the gcd of the valuations, not their minimum. The statement of part (b) asks only for independence of choices, which the exponent-gcd invariant proves; Theorem ?? gives the correct explicit formula.

Remark 11 (On the number of moves). *The bound $(k - 1) + \lfloor \log_2 P_0 \rfloor$ of Lemma ?? depends on the initial board through the initial product P_0 ; part (a) asks only for finiteness, which the bound supplies for every initial board. The bound is uniform in the choices of Confucius.*